AI, GenAI & Agentic AI for Automotive Leaders and Developers

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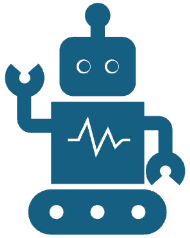
Day-Wise Outline

- Day 1: AI/ML/DL – Foundation for Automotive Innovation
- Day 2: AI Leadership, Risk Management & Storytelling with Data
- Day 3: Generative AI and Document Automation
- Day 4: Strategic Thinking – Competing in the AI Age
- Day 5: Agentic AI, Automation, Decision Making for Digital Teams



Agentic AI, Automation, and Decision Making for Digital Teams

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GKTCS Innovations

Objective

Explore Agentic AI,

prompt engineering,

chaining workflows, and

decision-making using dashboards.

Agenda

- Evolution of AI Automation
- Core Components of Agentic AI
- Reasoning Patterns in Agents
- Application Scenarios

Agenda

Prompt Engineering Essentials → Types of Prompts:

Zero-shot → Few-shot

Chain-of-Thought → Structured Prompts

Session 2

Automotive Agentic AI Use Case Demo

Use Case:

Mercedes-Maybach GLS 600 4MATIC Night Series

Problem

Manual extraction of automotive specifications

from PDFs, brochures, and pricelists

is time-consuming and error-prone.

Agentic Workflow Roles

Agent	Function
Data Extractor Agent	Extracts vehicle specifications (Torque, Engine, CO_2, Fuel Consumption, Dimensions, Price, Ambient Lights, Air Suspension, etc.)
Data Structurer Agent	Converts the extracted data into structured formats (JSON/CSV)
Decision Assistant Agent	Provides recommendations for product comparison, upselling, or variant suggestions

Demo

Script-based Extraction using

LLM Agents / LangChain / AutoGen

Fetch some of below features

- Torque/Engine capacity
- Dimensions
- Price
- Fuel consumption
- Fuel Type

Fetch some of below features

- Co2 emissions
- Weight
- Part Assist, Ambient Lights
- Front/Rear Air suspension

Sample Output

Torque: **770 Nm**

Engine: **3,982 cc V8**

CO_2 Emissions: **313 g/km**

Price: **£206,350**

Sample Output

Fuel: **Super, Mild Hybrid**

Suspension: **AIRMATIC (Front/Rear)**

Ambient Lighting: **64 Colors**

Session 3: Visual Storytelling & Decision Making with GenAI

- Why Visual Insights?
- Auto-Generated Insights
- RAG + LLM Chatbot
- Auto-Generated Dashboard

Session 4: Governance & AI Readiness Planning

- AI Maturity Model for Digital + Automotive Teams
- Building AI Roadmaps & KPIs
- Responsible AI & Compliance in product data automation

Checklist Activity

“Is Your Digital/Automotive Team AI-Ready?”

Complete self-assessment worksheet.

Agentic AI & Task Chaining

What is Agentic AI?

Core components:

Agents

Tools

Planner

Executor

What is Agentic AI?

- Agentic AI = Smart AI agents that
- Understand goals
- Plan steps
- Execute tasks
- Work together (like a team)

What is Agentic AI?

New way of using AI

to **plan, execute, and manage tasks**

independently

like an intelligent assistant

that thinks and acts.

What is Agentic AI?

ACT LIKE A RESPONSIBLE ASSISTANT,

DOING TASKS **INDEPENDENTLY**

BASED ON GOALS YOU GIVE IT.

Agentic AI & The Evolution of AI Automation

From Prompts

Tools

Agents

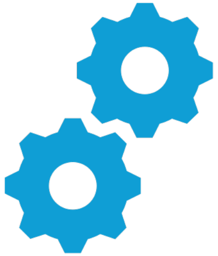
Introduction: Evolution of AI Automation

Phase	What It Means	Example
Prompts	AI responds to human instructions via natural language.	"Summarize this document."
Tools	AI integrates with APIs and tools to perform actions beyond text generation.	AI pulls data from a database and generates a report.
Agents	AI becomes autonomous, using reasoning and tools to complete multi-step tasks without constant human input.	AI monitors logs, detects incidents, creates tickets, and suggests fixes.

What is Agentic AI?

AI systems that

Act **autonomously**



Use **tools & APIs**

What is Agentic AI?

PERFORM

REASONING LOOPS

PLAN → ACT → OBSERVE → ITERATE

What is Agentic AI?

Collaborate with

other agents or humans

Have **memory &**

context awareness

Evolution Timeline

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Step 1: Prompts (LLM 1.0)

AI generates text based on user prompts

Limitation

Passive,

no tool execution

Step 2: Tools (LLM + Function Calling)

AI uses external tools & APIs

Examples:

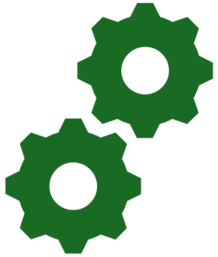
Query databases

Call external services

Limitation: Human still manages the sequence

Step 3: Agents (LLM 2.0 & Beyond)

AI autonomously achieves objectives



Performs multi-step workflows using tools

Uses memory & reasoning to adapt

Why Agentic AI?

Traditional Automation	Agentic AI
Predefined, rule-based	Adaptive & autonomous
Requires human orchestration	Self-managed workflows
Single tasks	Multi-step decision making

Use Case: IT Operations (AIOps)

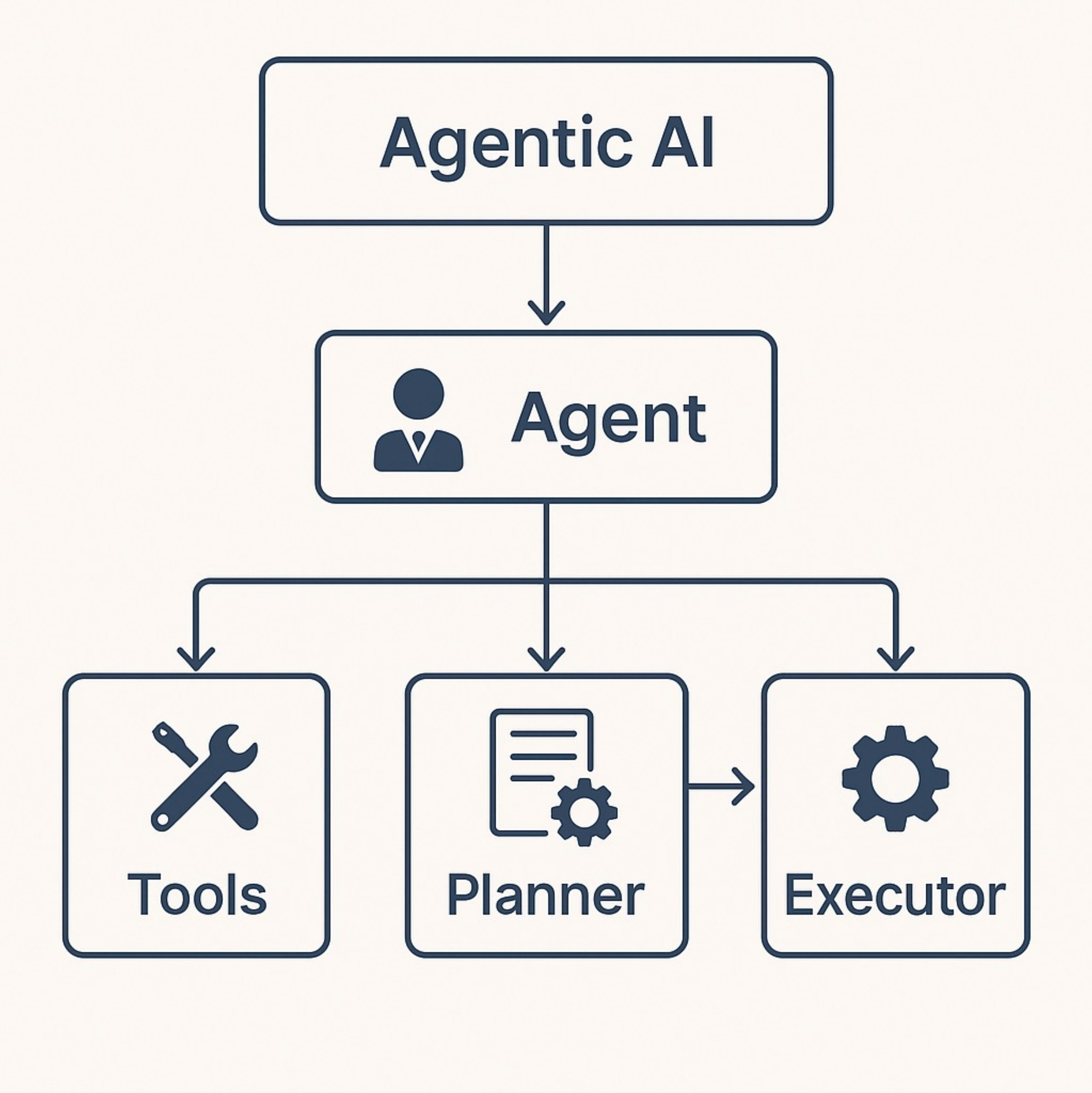
Stage	Example
Prompt	“Summarize system logs.”
Tool Use	GPT reads logs, uses Grafana API to send alerts.
Agentic AI	Monitors logs 24/7, detects issues, creates tickets, suggests fixes, and communicates via Slack or Teams.

Benefits of Agentic AI

- Reduces human intervention
- Handles complexity
- Learns and adapts over time
- Cross-domain workflow execution

Core Components of Agentic AI

1. Agents
2. Tools
3. Planner
4. Executor


Agentic AI
Agent
Tools
Planner
Executor

1. Agents

Individual units with specific roles or goals.

Can reason, learn, and make decisions

based on their goals.

2. Tools

Resources or capabilities available to agents.

For example: APIs, databases,

web search engines, or AI models.

3. Planner

Determines how to achieve the goal.

Decides which agents

should collaborate, and in what order.

4. Executor

Executes the planned tasks.

Coordinates interaction

between agents and tools

to complete tasks.

Core Components of Agentic AI

Component	Role	Purpose
Agent	Decision-making unit	Understands the goal, decides next action
Tool	Action executor	Interfaces with external systems (APIs, databases, scripts)
Planner	Task decomposer	Breaks down objectives into actionable steps
Memory	Context manager	Remembers previous actions, results, or conversation states
Executor	Task orchestrator	Executes tasks in order, manages agents, and loops

Real-World Analogy

Think of

Agentic AI system as a

Project manager + Technical team

Agent

MAKES DECISIONS

“WHAT DO I NEED TO DO NEXT?”

Tool

Executes the task

Call API,

run function,

fetch data

Planner

BREAKS BIG GOALS INTO SUBTASKS

STEP 1: COLLECT DATA,

STEP 2: ANALYZE

Memory

Remembers context/history

Last time,

this tool failed.

Try a different method.

Executor

Manages the workflow,

retries, loops

Coordinate all tasks in sequence

Automotive Agentic AI Workflow

Use Case Summary:

Extract, structure, analyze, and

suggest decisions based on

vehicle specs from brochures,

PDFs, or websites using Agentic AI.

Core Components Explained (Automotive)

Component	Role in Automotive Domain	Example (Mercedes-Maybach GLS 600 Data)
Agent	Decision-maker: Understands the goal, manages steps.	Agent decides: "Extract all key specs from brochure text and prepare comparison-ready data."
Tool	Performs actions: Uses OCR, LLM, web scraping, PDF parsing.	Use pdfplumber, GPT-4o, or BeautifulSoup to extract: - Torque: 770 Nm - Engine: 3,982 cc V8 - Price: £206,350

Core Components Explained (Automotive)

Component	Role in Automotive Domain	Example (Mercedes-Maybach GLS 600 Data)
Planner	**Breaks the goal into subtasks**: Defines extraction sequence.	Plan: 1. Extract performance specs 2. Extract comfort features 3. Extract pricing 4. Check for missing data
Memory	**Retains extracted context and history**: Stores intermediate outputs for consistency.	Memory stores: - "Engine capacity extracted from page 3" - "Suspension type already identified: AIRMATIC" - Logs features extracted for reuse

Core Components Explained (Automotive)

Component	Role in Automotive Domain	Example (Mercedes-Maybach GLS 600 Data)
Executor	**Orchestrates the workflow**: Runs tasks in sequence, loops for completeness.	Executor calls agents/tools: - Run PDF extraction - Call LLM summarization - Generate comparison or upsell recommendations

Benefits of Agentic AI in Automotive

- Automation of manual spec extraction
- Faster generation of sales materials & reports
- Reduced human error in vehicle data processing
- Supports real-time comparison & decision support

Reasoning Patterns in AI Agents

- ReAct
- Chain-of-Thought
- Plan-and-Execute

Why Reasoning Patterns?

Help	Help agents handle complex tasks
Enable	Enable multi-step decision making
Improve	Improve explainability and action execution

ReAct (Reasoning + Action Loop)

An agent **thinks step-by-step**,

decides what to do,

takes action using tools,

observes the result,

then **thinks again**.

This loop continues until the task is complete

Key Features

Interleaves **thinking and acting**

Handles **dynamic decision-making**

Useful for **real-time problem solving**

Example Flow

Agent: "What’s the torque of this car?"

→ Think: "I need to search for the data."

→ Act: Use a search API.

→ Think: "Now compare it to competitor data."

→ Act: Use comparison tool.

Chain-of-Thought (CoT)

The agent is encouraged

to **explain its reasoning process**

before giving an answer.

It doesn’t act with tools

but reasons clearly.

What is Chain-of-Thought (CoT)?

Instead of jumping to the final answer,

the model is instructed to **think aloud**

by breaking the problem into

logical reasoning steps.

Chain-of-Thought (CoT) Prompting

Forces the agent

to **explicitly write out**

its reasoning process

step-by-step before answering.

Chain-of-Thought (CoT) Prompting

This makes the AI's logic

transparent,

explainable, and

less error-prone.

Key Features

Enhances **clarity of reasoning**

Good for **complex calculations,**

comparisons, or multi-step logic

Reduces **LLM hallucination**

Example Flow

Step 1: The Maybach GLS has 770 Nm torque.

Step 2: The BMW X7 has 750 Nm torque.

Step 3: 770 > 750 → Maybach has better torque.

Final Answer: Maybach is superior in torque.

What is Chain-of-Thought (CoT)?

Improves accuracy, interpretability

often factual correctness,

especially for complex or

multi-step questions.

Plan-and-Execute (PnE)

The agent **first plans**

the full sequence of tasks,

then executes each step.

What is Plan-and-Execute (PnE)?

Powerful agent pattern that:

First plans a sequence of subtasks

based on the user request.

Then **executes each step**,

often using tools, APIs, or functions.

Plan-and-Execute

Separates

the **planning phase**

from the **execution phase.**

Plan-and-Execute

First, the agent **creates a full plan** (list of steps).

Then, it **executes each step sequentially,**

often using tools or APIs.

Key Features

Useful for **long-horizon tasks**

Provides **structured control**

Helps with **multi-stage workflows**

Example Flow

Agent Goal:

Generate a marketing brochure

from vehicle data.

Plan

Extract	Extract specs from brochure.
Compare	Compare to competitor models.
Write	Write upselling points.
Format	Format content into brochure template.

Execute

- Step 1 → Done.
- Step 2 → Done.
- Step 3 → Done.
- Step 4 → Done.

What is Plan-and-Execute (PnE)?

This improves

reliability,

transparency, and

modularity in AI task solving.

Summary Table

Pattern	Purpose	Best For
ReAct	Think + Act in loops	Real-time, tool-based reasoning
Chain-of-Thought	Step-by-step explanation	Transparent logic, QA systems
Plan-and-Execute	Plan first, then act	Complex workflows, pipelines

Application Scenarios

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GKTCS Innovations

Enterprise Apps

- Automated report generation
- Financial anomaly detection
- IT operations automation (AIOps)
- Customer support chatbots
- Workflow orchestration in CRM/ERP

Automotive Use Cases

- Vehicle brochure data extraction
- Real-time maintenance decision support
- Competitive feature comparison
- Predictive maintenance pipeline
- Sales team enablement through AI assistants

Conclusion

Agentic AI uses reasoning patterns to:

Solve complex tasks

Improve accuracy and transparency

Automate multi-step decisions

Prompt Engineering Essentials

Why Prompt Engineering?

The practice of designing effective inputs

to guide LLMs toward **accurate,**

contextual, and reliable outputs.

Why Prompt Engineering?

IN THE **AUTOMOTIVE DOMAIN,**

PROMPTS CAN BE DESIGNED TO:

DIAGNOSE VEHICLE ISSUES

RECOMMEND ACTIONS

EXTRACT STRUCTURED DATA FROM MANUALS OR LOGS

Types of Prompts with Automotive

Prompt Type	Definition	Automotive Example: Diagnostic Assistant
Zero-shot	Provide no examples; rely solely on task description	Prompt: \n"Diagnose the issue based on this error: Engine temperature is rising rapidly despite normal coolant levels."

Types of Prompts with Automotive

Prompt Type	Definition	Automotive Example: Diagnostic Assistant
Few-shot	Provide **2-5 examples** to help the model learn the pattern	**Prompt:** \n"Example 1: Input: Engine makes clicking sound → Output: Battery issue likely.\nExample 2: Input: Brakes squealing → Output: Brake pad wear. \nNew Input: Engine temperature rising despite coolant levels → Output: Possible thermostat malfunction."

Types of Prompts with Automotive

Prompt Type	Definition	Automotive Example: Diagnostic Assistant
Chain-of-Thought (CoT)	Ask the model to **reason step by step** before answering	**Prompt:** \n"Diagnose this issue: Engine temperature is rising rapidly. \nStep 1: Check coolant level → Coolant is normal. \nStep 2: Check thermostat functionality → Possible thermostat stuck closed. \nFinal Answer: Thermostat malfunction."

Types of Prompts with Automotive

Prompt Type	Definition	Automotive Example: Diagnostic Assistant
Structured Prompts	Use **JSON, XML, or predefined formats** for consistent output	**Prompt:** \n"Given the following vehicle symptom, output the diagnosis in JSON: \nInput: Engine temperature is rising. \nOutput Format: {"Symptom": "Engine Overheat", "Possible Cause": "Thermostat Failure", "Recommended Action": "Check thermostat and coolant system"}"

Types of Prompts with Automotive

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Summary of When to Use Each

Type	When to Use
Zero-shot	Simple, general tasks where LLM is pretrained
Few-shot	Custom tasks needing example-based learning
CoT	Complex decisions requiring reasoning
Structured	Output needs to be machine-readable (API integration, dashboards)

Real-World Automotive Applications

Vehicle Health Reports

Workshop Support Chatbots

Remote Diagnostics for Connected Cars

Owner Manual Query Assistants

Session 3: Visual Storytelling & Decision Making with GenAI

- Why Visual Insights?
- Auto-Generated Insights
- RAG + LLM Chatbot
- Auto-Generated Dashboard

Why Visual Insights?

Leaders prefer **charts + narratives** over raw tables

Visuals enable **faster, clearer decision-making**

Context + visualization = **actionable insights**

The Problem

Traditional Approach	Challenges
Raw vehicle data tables	Hard to interpret quickly
Service logs / sales sheets	Time-consuming for execs

GenAI Solution with LangGraph

Automate visual + narrative insights from raw data

LangGraph Workflow

Ingest	Ingest raw data (sales, CO_2, features)
Generate	Generate charts automatically
Summarize	Summarize insights with GPT-4o
Output	Output executive-ready reports

LangGraph Architecture

[Raw Data Input]

↓

[Chart Generator Node] → Matplotlib/Plotly Visuals

↓

[LLM Summarizer Node] → GPT-4o Narrative Insights

↓

[Report Generator Node] → PDF / PPT Report Output

Automotive Example Use Case

INPUT: VEHICLE SALES + CO_2 DATA

OUTPUT:

CHARTS: SALES VS EMISSIONS TREND

NARRATIVE:
"SUV SEGMENT LEADS SALES BUT EMISSIONS UP 15%. RECOMMEND HYBRID PROMOTION IN Q4."

Benefits of LangGraph Approach

Visual & Narrative Automation

Faster Leadership Decisions

Reusable AI Pipelines

Scalable for Automotive Data

Automotive Applications

- Executive Briefings
- CO_2 Compliance Dashboards
- Vehicle Feature Comparisons
- Sales Strategy Recommendations

Auto-Generated Insights

Use extracted specs

for product comparisons or

sales insights.

Use Case Overview

Goal:

Automatically extract specs from vehicle brochures and generate:

Product Comparison Reports

Sales Insights or Upselling Recommendations

Governance & AI Readiness Planning

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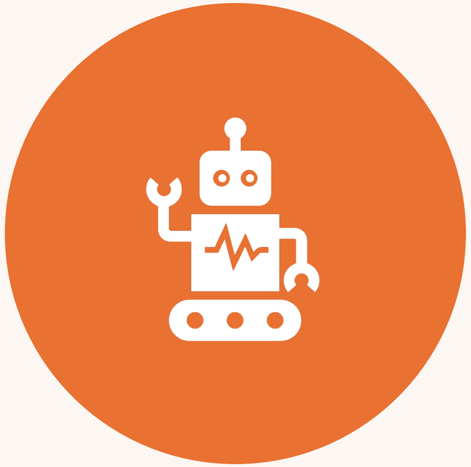
Checklist Activity



“IS YOUR DIGITAL TEAM AI-READY?”

WORKSHEET AND TEAM RATING

AI Maturity Model: Overview

Outlines

how prepared and capable

an organization is to adopt,

scale, and govern AI initiatives.

AI Maturity Model: Overview

Typically progresses through

5 stages across

3 key pillars

AI Maturity Model: Overview

Pillar	Focus Area
Tools	Tech infrastructure, platforms, data pipelines
People	Skills, roles, leadership support
Processes	Governance, risk management, lifecycle controls

Maturity Levels (1–5)

Level	Description	Characteristics
1. Initial	Ad hoc or pilot usage	Uncoordinated tools, siloed data, limited awareness
2. Emerging	Some tools, isolated teams	Early-stage adoption, basic POCs, limited governance
3. Established	Centralized AI team, aligned with business	Basic governance, repeatable processes, growing toolset
4. Integrated	Cross-functional integration	AI embedded into business processes, role-based access, model monitoring
5. Optimized	AI as strategic differentiator	Full lifecycle governance, explainability, responsible AI practices in place

Pillar-Wise Breakdown with Examples

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1. Tools (Technology & Platforms)

Maturity	Example Tools	AI Readiness Indicators
Level 1	Excel, manual data prep	No AI platform, no versioning
Level 2	Jupyter, Power BI, OpenAI trials	Early experimentation, lack of APIs
Level 3	Azure ML, DataBricks, HuggingFace	MLOps pipelines, central AI environments
Level 4	RAG stack, LangChain, Model monitoring tools (MLflow)	Observability, secured endpoints
Level 5	Custom GenAI platform + API Gateway + Vault	Secure, scalable, and reusable AI assets

🔧 1. Tools (Technology & Platforms)

Governance Fit:

Introduce tools that support logging,

access control, explainability

(e.g., Azure OpenAI with logging + audit trails).

👥 2. People (Skills & Organizational Support)

Maturity	Roles Present	Readiness Indicators
Level 1	Individual data analysts	No AI strategy, no formal training
Level 2	ML enthusiasts, IT support	POC-level work, unclear responsibilities
Level 3	Data Scientists, Product Owners	AI team in place, early skill-building
Level 4	AI PMs, Security Leads, Prompt Engineers	Cross-functional collaboration
Level 5	AI Governance Council, AI Ethics Officer	Continuous upskilling, center of excellence (CoE)

Governance Fit:

Assign **Data Steward,**

AI Risk Owner, and

Model Reviewer roles.

⚙️ 3. Processes (Governance, Risk, Lifecycle)

Maturity	AI Governance Process	Readiness Indicators
Level 1	None	No model validation or security reviews
Level 2	Informal reviews	Ad hoc documentation, no audit trails
Level 3	Basic governance policies	Risk register, initial compliance reviews
Level 4	Responsible AI framework	Role-based access, explainability, data lineage
Level 5	Auditable, ethical AI pipeline	ISO/IEC 42001 readiness, real-time monitoring, SLA-bound AI ops

Governance Fit: Implement

Model documentation (Model cards)

Prompt injection testing

Data privacy reviews

Usage logs + alerting system

Example AI Maturity Use Case

GenAI-Powered Employee Assistant (Live in HR Portal)

Example AI Maturity Use Case

GenAI-Powered Employee Assistant (Live in HR Portal)

Pillar	Current State (Level 3)	Next Step (Level 4 Goal)
Tools	Azure OpenAI + RAG + ChromaDB	Add Observability (Prometheus), Vault Integration
People	Data scientist + HR Analyst	Add Prompt Engineer + AI Risk Owner
Processes	Manual review + policy doc	Add audit trail, moderation filter, ISO-based policy checklist

What You Can Do as a PM

Run a **Maturity Assessment Workshop**

with your stakeholders

Assign owners for tools,

people, and processes

What You Can Do as a PM

Build an **AI Readiness Roadmap**

aligned to maturity gaps

Start with 1–2 **governance-enabling pilots**

(e.g., secure RAG chatbot)

Building an AI adoption roadmap (6-pillar model)

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🧭 AI Adoption Roadmap: 6-Pillar Model

Pillar	Focus Area	Objective
1. Strategy & Vision	Align AI with business goals	Set direction and leadership buy-in
2. Data Readiness	Ensure clean, secure, accessible data	Power AI with quality input
3. Technology & Tools	Build scalable AI infrastructure	Choose the right platforms & stacks
4. People & Skills	Enable talent and org structure	Upskill team for AI roles
5. Governance & Risk	Establish ethical and secure use	Comply with regulations, prevent misuse
6. Use Cases & Value	Deliver high-impact pilots	Show value through real AI solutions

1. Strategy & Vision

Goal:

Define **why** and **where**

AI adds value across business units.

Example

"Implement GenAI-powered

HR support to reduce ticket volume

by 50% in 6 months."

Governance Tie-In

Set AI policy scope

(who owns AI decisions?)

Establish AI steering committee

2. Data Readiness

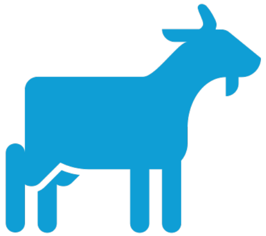
Goal: Ensure that data is **available,**

clean, secured, and **compliant**.

Example

Build a unified data lake from

HRMS, CRM, and knowledge bases



for RAG-based AI assistant.

2. Data Readiness

GOVERNANCE TIE-IN:

ENSURE GDPR/DPDP COMPLIANCE

IMPLEMENT **DATA ACCESS CONTROL,**

LINEAGE, AND **RETENTION POLICIES**

ASSIGN **DATA STEWARDS**

3. Technology & Tools

Goal:

Adopt and integrate the right

AI platforms,

APIs, and infrastructure.

Example:

Use Azure OpenAI + LangChain

+ FAISS for internal GenAI chatbot

Secure with API Gateway + Vault + OIDC login

Governance Tie-In

Only allow secure, observable APIs

Log and monitor all model interactions

Use RAG to control LLM outputs from verified sources

4. People & Skills

🎯 **Goal:**

Upskill workforce and

assign critical AI governance roles.

💼 Example

Train Devs on prompt engineering,

PMs on Responsible AI,

create **AI Risk Owner** role.

Governance Tie-In

Define roles like:

AI PM

Model Reviewer

Ethics Officer

Build awareness around bias, misuse, data protection

5. Governance & Risk Management

Goal:

Create policies,

frameworks, and

audits for safe, ethical AI.

Example

Define a checklist to evaluate new GenAI features:

☐ Prompt injection tested

☐ Logging enabled

☐ No sensitive data exposure

Key Areas

- ISO/IEC 42001 compliance readiness
- Implement Responsible AI practices
- Use audit trails, explainability, and human oversight

6. Use Cases & Value Realization

Goal:

Identify, prioritize, and

implement **high-impact,**

low-risk pilots.

Example

- Pilot GenAI for:
  - HR ticketing
  - Sales email drafts
  - IT troubleshooting

Track ROI

TIME SAVED,

ACCURACY,

TICKET REDUCTION

Governance Tie-In

Only deploy GenAI pilots

with usage monitoring and access restrictions

Retire or improve features

that don’t meet compliance or ethical thresholds

Visual Summary

AI Adoption Roadmap (6-Pillar)

Sample Milestones (Timeline-based View)

Phase	Key Activities
Month 1–2	Set strategy, assess data, form AI Council
Month 3–4	Implement secure AI stack, define use cases
Month 5–6	Launch pilots with audit & monitoring
Month 7–8	Evaluate, scale success, document policies

Deliverables You Can Create

- AI Strategy Document
- Data Readiness Checklist
- Governance Policy + Role Matrix
- Secure GenAI Pilot Plan
- Risk Register and Maturity Dashboard

Final Thought

A successful AI roadmap isn’t just about technology

it’s about aligning vision, securing data,

enabling people, and governing

every step responsibly.

Defining KPIs for AI projects

Critical to track success,

justify investment, and

ensure alignment

with governance and

AI readiness goals.

Why KPIs for AI Projects Matter?

Measure value delivered by AI

(not just tech progress)

Track risks and compliance

(bias, data privacy, auditability)

Why KPIs for AI Projects Matter?

Ensure continuous improvement

(performance, accuracy, efficiency)

Align teams and stakeholders

with clear success outcomes

✅ KPI Framework for AI Projects

Category	Focus	Examples
1. Business Impact	ROI, efficiency, productivity	Cost saved, time saved
2. Model Performance	Accuracy, bias, drift	F1 score, hallucination rate
3. Governance & Risk	Compliance, audit, explainability	% prompts logged, data access violations
4. Adoption & Usability	Engagement, feedback, usability	Active users, satisfaction score

Examples of KPIs by AI Use Case

GenAI-Powered HR Assistant

GenAI-Powered HR Assistant

KPI Type	KPI Example	Description
Business Impact	🔷 40% reduction in HR support tickets	Measures cost/time saving
Model Performance	🔷 90% accurate retrieval from policy docs	Precision in RAG queries
Governance	🔷 100% prompts logged with user ID	Audit trail completeness
Usability	🔷 4.5/5 average user rating	User feedback loop

2. AI for Sales Email Generation

KPI Type	KPI Example
Business Impact	🔷 25% increase in response rate
Model Performance	🔷 <5% flagged for incorrect personalization
Governance	🔷 Zero PII used in prompt data
Usability	🔷 70% of sales team using weekly

3. Document Summarization Bot

KPI Type	KPI Example
Business Impact	🔷 3x faster document review time
Model Performance	🔷 Summarization accuracy score ≥85%
Governance	🔷 All sources cited in output
Risk	🔷 Hallucination rate <3% (based on QA review)

🛡️ Governance-Aligned KPIs

Governance Domain	KPI Example
🔐 Data Privacy	% of prompts that exclude PII
📊 Auditability	% of interactions with complete logs
🧠 Explainability	% of outputs with source or justification
⚠️ Risk Detection	# of prompt injection attempts blocked
✅ Compliance	ISO/IEC 42001 control coverage %

🎯 Setting SMART KPIs

Specific: “Reduce average response time from 10s to 3s”

Measurable: “Flag rate < 5% for toxic outputs”

Achievable: “Train 20 internal users in Prompt Engineering”

Relevant: “Improve model explainability score for auditors”

Time-bound: “Achieve 80% adoption in 3 months”

Summary Table

Category	KPI Examples
Business Value	Cost savings, process automation %
Model Accuracy	Precision/Recall, hallucination %
Compliance	Prompt audit coverage, GDPR incidents
Security	Tokens accessed via Vault only %
User Experience	Adoption rate, satisfaction score

Team roles: Product, DevOps, Security, QA, Data

Function	Role in AI Lifecycle	Key Governance Contribution
🔷 Product	Define value, scope, ethics	Align use cases with business & compliance
🔷 DevOps	Deploy & monitor models	Secure, scalable, auditable infrastructure
🔷 Security	Protect systems & data	Enforce privacy, access control, compliance
🔷 QA	Test accuracy & safety	Validate fairness, bias, prompt injection
🔷 Data	Prepare & govern data	Ensure data quality, lineage, policy compliance

AI Readiness Checklist Activity

As a Project Manager of a Digital Team,

conducting an AI Readiness Checklist Activity

is a valuable way to assess your team’s

current capabilities, gaps, and governance maturity

before deploying AI/GenAI initiatives.

Is Your Digital Team AI-Ready?

OBJECTIVE:

HELP EACH FUNCTION

PRODUCT, DEVOPS, SECURITY, QA, DATA

ASSESS AI READINESS FROM A GOVERNANCE AND

CAPABILITY STANDPOINT.

Section 1: Team-Wise AI Readiness Checklist

Area	Readiness Question	Example Indicator	Yes / No / Partial
Product	Have we identified high-impact, low-risk AI use cases?	GenAI for HR, IT Helpdesk	✅ / ❌ / ⚠️
Product	Do we have a clear AI success metric (e.g., ROI, adoption)?	Target: 40% reduction in ticket load	✅ / ❌ / ⚠️

Section 1: Team-Wise AI Readiness Checklist

Area	Readiness Question	Example Indicator	Yes / No / Partial
DevOps	Is our infrastructure ready to deploy and monitor GenAI apps?	Azure + API Gateway + Vault	✅ / ❌ / ⚠️
DevOps	Are logs, usage metrics, and observability dashboards in place?	Prometheus, Grafana	✅ / ❌ / ⚠️

Section 1: Team-Wise AI Readiness Checklist

Area	Readiness Question	Example Indicator	Yes / No / Partial
Security	Have we implemented access controls (RBAC, OIDC)?	User roles enforced via identity provider	✅ / ❌ / ⚠️
Security	Is prompt injection mitigation in place?	Input filters, jailbreaking tests	✅ / ❌ / ⚠️

Section 1: Team-Wise AI Readiness Checklist

Area	Readiness Question	Example Indicator	Yes / No / Partial
QA	Are we testing GenAI for bias, hallucination, and safety?	Prompt test suite maintained	✅ / ❌ / ⚠️
QA	Do we have test cases for prompt override attempts?	“Ignore all instructions” blocked	✅ / ❌ / ⚠️

Section 1: Team-Wise AI Readiness Checklist

Area	Readiness Question	Example Indicator	Yes / No / Partial
Data	Is our data catalog tagged, clean, and compliant?	Metadata for access scope in place	✅ / ❌ / ⚠️
Data	Do we have a process for data source approvals?	Stewards review before ingestion	✅ / ❌ / ⚠️

Section 2: Team Self-Rating (Score from 1 to 5)

Team	1 = Not Ready	5 = Fully Ready
Product	?	Example: 4 – Identified use cases, need risk scoring
DevOps	?	Example: 3 – Deployed model but observability missing
Security	?	Example: 2 – RBAC in place but prompt filters missing
QA	?	Example: 3 – Testing basic inputs, no edge-case coverage
Data	?	Example: 2 – Data tagged but not governed

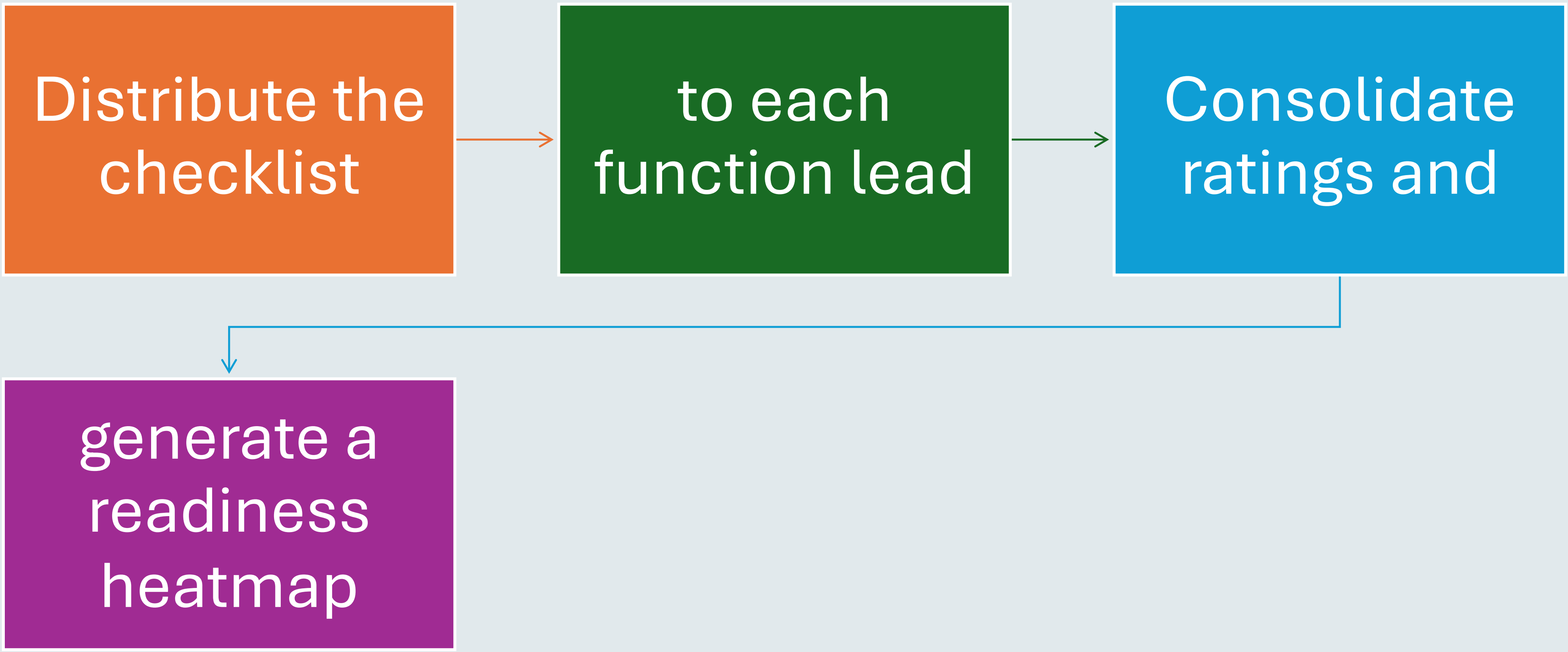
Example Output Summary

Team	Rating (1–5)	Readiness Comment
Product	4	Good use case selection, needs ethics checklist
DevOps	3	Infra in place, but lacks logging & alerting
Security	2	RBAC done, prompt abuse controls pending
QA	3	Manual checks in place, automated test plan WIP
Data	2	Raw data used; need stewards and source control

Governance Insights from Activity

If You Score Low On...	Recommendation
Security (≤2)	Set up threat model, RBAC, prompt filter rules
Data (≤2)	Tag data, define access rules, assign data stewards
Product (≤3)	Create Responsible AI charter + ROI KPIs
QA (≤3)	Build GenAI QA test plan: bias, hallucination, toxicity
DevOps (≤3)	Add observability, logging, and audit trails

Next Steps for You as PM



Next Steps for You as PM

Prioritize initiatives to close the gaps

(e.g., training, policies, tooling)

Present findings in your

AI Steering Committee or leadership review

Happy Learning !!
Thanks for Your Patience ☺

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GKTCS Innovations